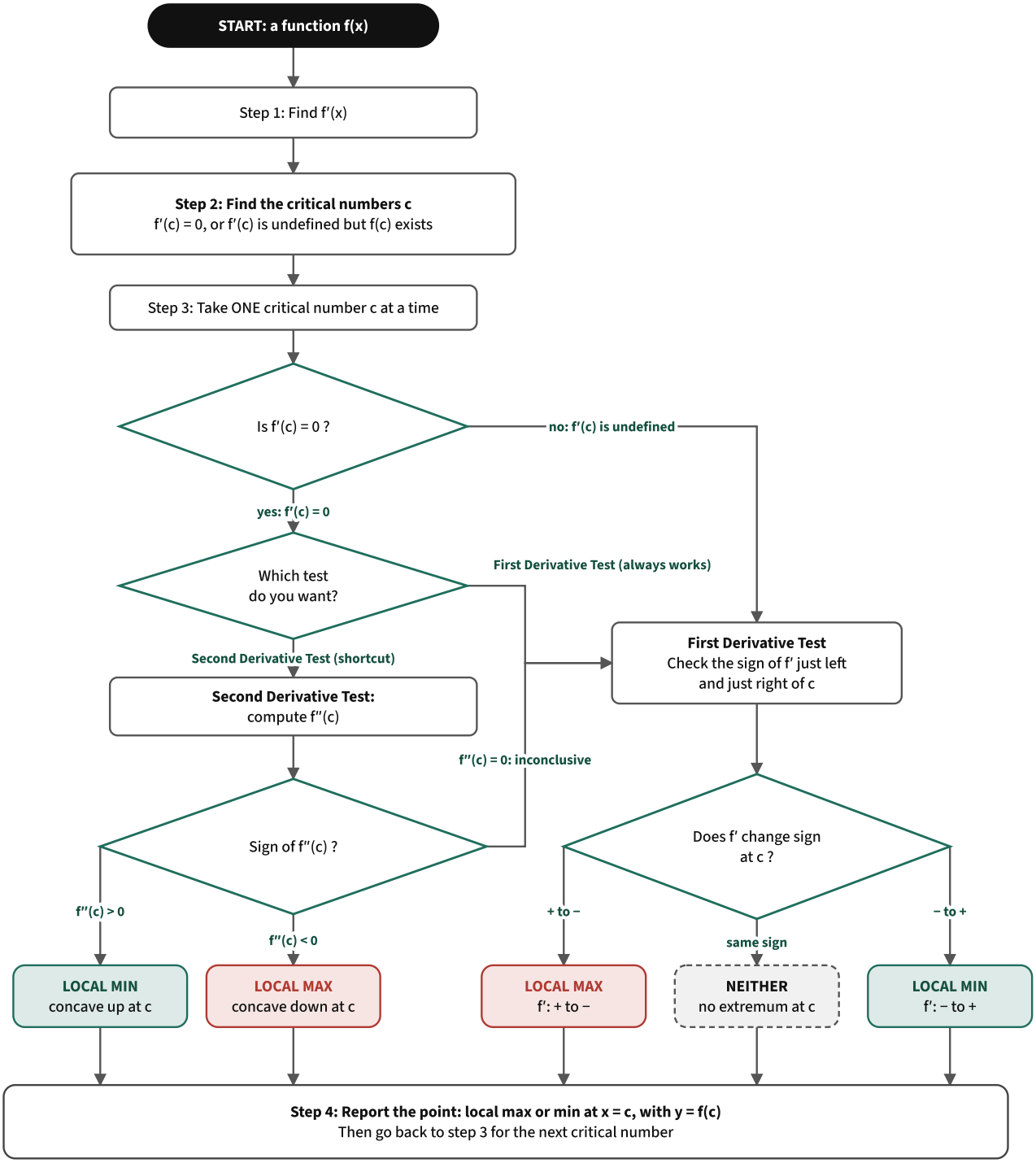
Local extrema: First & Second Derivative Tests

Work through every critical number c, one at a time.



Critical numbers must be in the domain

If f(c) does not exist, c is not a critical number, even if f′ is undefined there.

f″(c) = 0 does not mean "neither"

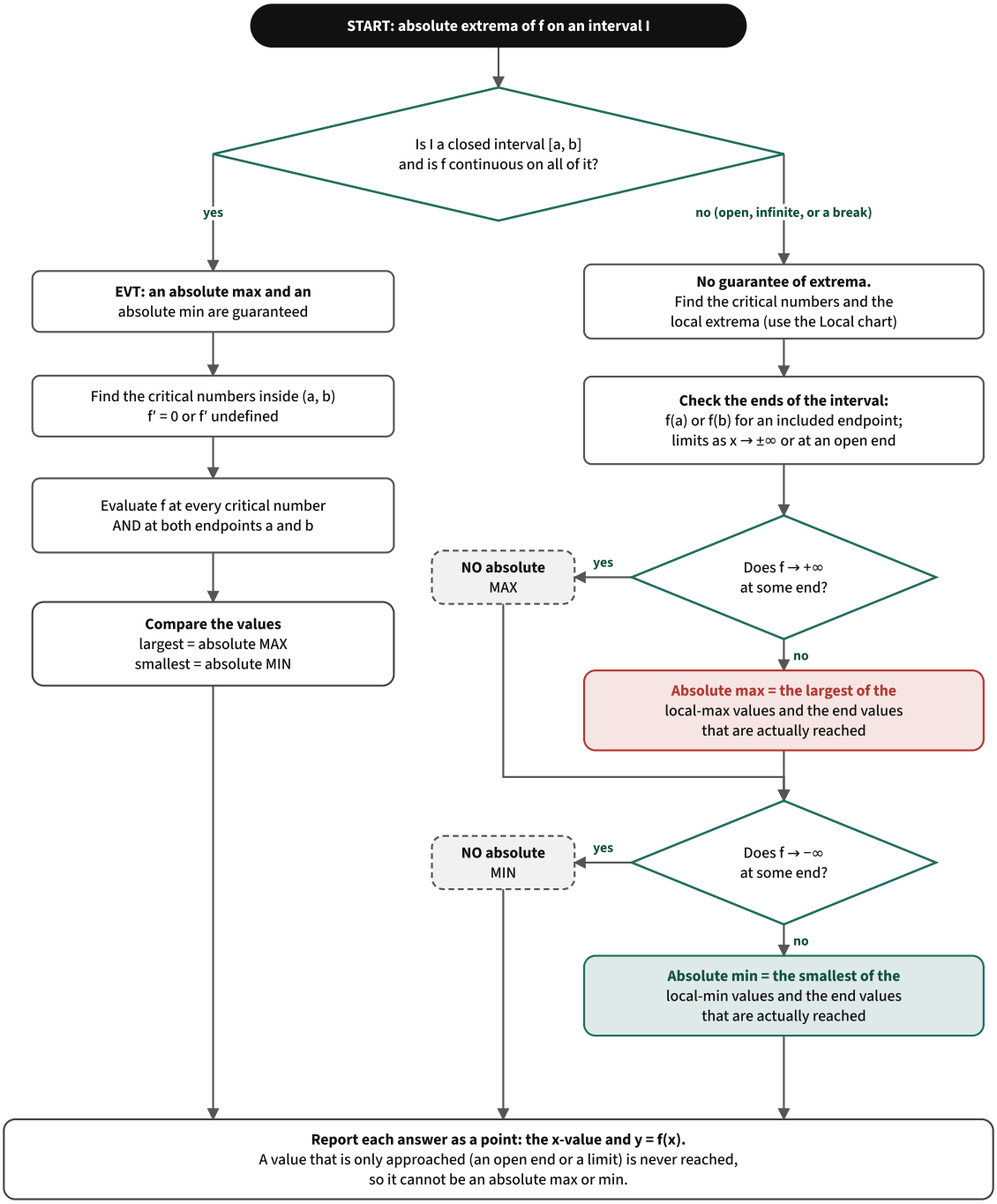
The Second Derivative Test is just silent. Go to the First Derivative Test and check the sign change.

The answer is a point

The critical number c is the x-value. Plug it into f to get the y-value of the local max or min.

Global (absolute) extrema

Start by asking whether the interval is closed and f is continuous on it.



Shortcut on an open or infinite interval

If f has exactly one critical number c on the interval, and f′ changes from − to + there (or f″ > 0 everywhere), c gives the absolute *min*. From + to − (or f″ < 0 everywhere), c gives the absolute *max*.

Endpoints only count if they are included

A square bracket means the endpoint is in the interval and must be tested. A round bracket or ∞ means the value is only approached, never reached.

Absolute extremum ≠ local extremum

An absolute max is the highest value on the whole interval. It can be a local max or an endpoint, and a local max does not have to be the absolute max.